

LPW_CONSTANT v2 — corrected successor (Return-05 Priority 1)

Record: LPW-CONSTANT-V2-20260913. **Role:** corrected constant evaluation layer on top of the endorsed existential proof (six interfaces PASS WITHIN SCOPE) and v1 (frozen). **Script:** lpw_constant_v2.py (fail-closed `ck()`→`SystemExit`, no bare asserts, deterministic, both python and python -O byte-identical program output; exit receipts in `exit_normal.txt`/`exit_O.txt`).

1. The two v1 defects and their repairs

D1 — the published decimal did not follow from the exact fraction. v1 used the integer bound $B_{3,\text{rat}} = 3790446482793$, whose exact coefficient is $260/(3790446482793 \cdot 2 \cdot 10^{21}) = 6.238542702935588\text{e-}44$, but displayed $6.239277637492383\text{e-}44$ (computed from the ROUNDED denominator $3.79\text{e}12 = 3,790,000,000,000$). The published decimal was ABOVE the exact fraction. v2 derives the published decimal from the exact integer it actually uses, with a hard guard published exact fraction (and a power-of-ten floor demonstrated by exact rational comparison).

D2 — the fourth-moment identity was false. v1 used $E[(|X|+|Y|)] = 12 + 16/$. The correct value is $12 + 32/$ ($E|X| = \sqrt{2/}$, $E|X|^2 = 1$, $E|X|^3 = 2\sqrt{2/}$, $E|X| = 3: 2 \cdot 3 + 8 \cdot (2\sqrt{2/})(\sqrt{2/}) + 6 = 12 + 32/$ 22.186, not 17.093). v2 adopts the author-side amplitude majorant (Return-05's repair lemma): each Fourier mode's derivative contribution is bounded by its frequency factor times the amplitude $R = \sqrt{X^2 + Y^2}$, with $E[R] = 8$ exact ($E[X + 2X^2Y^2 + Y] = 3 + 2 + 3$) and $E[R] = \sqrt{E[R^2]} = \sqrt{2}$. Then, with the same spectral sums,

$$E\|f\|_{\{C^3\}^4} \leq 8 \cdot S_3^4, \quad E\|f\|_{\{C^4\}} \leq \sqrt{2} \cdot S_4.$$

The repair's multiplier (8) is SMALLER than v1's false one (17.09), so v1's caps survive conservatively (checked in-script), and v2 publishes the improved caps the repair yields. No silent relabeling: the false identity is retired, and a negative test (`M_WRONG_MOMENT`) rejects it.

2. Certified headline (exact Fractions / outward enclosures)

For the exact two-dimensional side-24 six-pin typed maximum-saddle pair-Palm law of the pinned candidate (cf58f72e...e1a5), under v1's unchanged uniform inputs (density floor $m = 10^{-21}$ on `JBOX(R)`, covariance floor $= 0.123809288439$ on $[0, R]$ with $R = 1\text{e-}5$, regression-operator bounds $A_3 = 71.10596$, $A_4 = 262.81357$, Isserlis $\sup E\|U_r\| = 38.23$, $\sup|u_r| = 1.28$, $\sup E\|V10_r\| = 3.175$, $\sup\|(u_r, j)\| = 2.3701308$):

$$\begin{aligned} B_3 &\leq [1 + 8^{\{1/4\}} \cdot S_3 + A_3(38.23^{\{1/4\}} + 1.28)]^4 = 2.082\text{e}12 & (B_{3,\text{rat}} = 2082000000000) \\ B_4 &\leq \sqrt{2} \cdot S_4 + A_4(3.175 + 2.3701308) \approx 4344.9 & (K_{\text{rat}} = 8690) \\ 1 - q(r, 6/5) &\geq c \cdot r^3 & \text{for all } 0 < r \leq r_0, \\ c &= 260 / (2082000000000 \cdot 2^{40} \cdot 10^{21}) = 1.1357762846347807215698804\text{e-}43 \\ &\geq 1.13\text{e-}43 & \text{(downward-safe display)} \\ &\geq 10^{-43} & \text{(power-of-ten floor by exact rational comparison)} \\ r_0 &= 1/2224640 = 1/(256 \cdot 8690) \approx 4.495109\text{e-}7 > 0. \end{aligned}$$

Hard checks (exact Fractions): $16 + 8Kr = 1/64 + 1/32 = 3/64 < 1/16$; $256 \cdot K \cdot r = 1$ exactly; path clearance $103/3840 > 0$; weight quotient $65 \cdot 16 = 1040 = 4 \cdot 260$; exponent $1 + 4 - 2 = 3$; thin-box

volume 32 r. Spectral sums $S_3 = 554.3913563$, $S_4 = 2041.832733$ (L2 frequency norm) recomputed to v1's certified precision; v1's caps $B_3 = 3790446482793$ and $B_4 = 4715.749569$ both survive the repair (verified).

3. Relationship to v1 and to the author-side fallback

- v1's headline ($c = 6.239e-44$, $r = 1/2414592$) is FROZEN as a historical artifact with its two defects recorded (D1, D2); its caps remain valid under the repair (smaller multipliers), so no downstream user of v1's caps is misled — but v1's displayed decimal and its moment derivation are wrong as written, and v2 is the current corrected version.
- v2 is STRONGER than v1's true fraction on both coordinates ($1.136e-43$ vs $6.239e-44$; $1/2224640$ $4.495e-7$ vs $1/2414592 = 4.141e-7$) because the amplitude majorant is smaller than the false moment multiplier.
- The author-side deliberate fallback ($r_* = 10^{-28}$, $c_* = 10^{-1235}$, analytically verified) is a separate route; it adds no radius coverage beyond $(0, r]$ and is retained as a cross-justification.

4. Negative tests (mutation suite; each injected via MUTATION=, rejected with exit 1)

- M_WRONG_MOMENT: injecting $E[(|X|+|Y|)] = 12 + 16/$ is detected (the correct $12 + 32/$ is enforced) — exit 1.
- M_DECIMAL_ABOVE_EXACT: publishing a decimal above the exact fraction is detected by the guard published exact — exit 1.
- eigenfloor_up (upward-rounded eigenfloor 0.126553449817289) — exit 1.
- hessian_power (r^3 substituted for the r pair weight) — exit 1.
- box_width (16 in place of 32 r) — exit 1.
- planar_moments ($1 - a = 0$ in place of the exact $9.65254179896e-123 > 0$) — exit 1. All six reject in both modes. The clean run exits 0 with CERTIFICATE: PASS; program output is byte-identical across python and python -O.

5. Remainder ledger

item	treatment
S_3, S_4 (554.3913563, 2041.832733)	v1-certified sums (L2 frequency norm, certified 2D shell tails $4.07e-32 / 1.11e-30 + \text{ROUND}$), recomputed here to displayed precision; used as exact Fractions at displayed precision
amplitude majorant ($E[R] = 8$, $E[R] = \sqrt{2}$)	exact; $E[R] = E[X + 2X^2Y^2 + Y] = 3 + 2 + 3$
$m = 10^{-21}$, $\epsilon = 0.123809288439$	v1's certified floor (contract boxed formula, changed compact set $J\text{BOX}(R)$, interval modulus — unchanged inputs)
A_3, A_4 , Isserlis sups, sup norms	v1's values, unchanged (independent of the defect)
$B_3, \text{rat} = 2082000000000$, $K_rat = 8690$	integer ceilings above the repaired mpf bounds
c	exact Fraction from the integer denominator; downward-safe display; power-of-ten floor by exact comparison

Table 1 – continued

item	treatment
$r = 1/2224640$	exact rational (K-term binds: $1/(256 \cdot 8690) < R = 1e-5$)

6. Falsifier

An interval-rigorous recomputation producing, on the same domain and law, a spectral sum S_3 or S_4 below the displayed values beyond their certified tails, an amplitude-moment $E[R] \leq 8$ or $E[R] > \sqrt{2}$, a repaired B_3/B_4 exceeding the published integer ceilings, or a final coefficient below the exact fraction $260/(2082000000000 \cdot 2^{-10^{21}})$, kills the corresponding input and, if load-bearing, the corrected pair. The mutation suite must continue to reject every injected defect (rerun with `MUTATION=`; any zero exit fails this report).

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